Mock 1 Medium 01 | Retail Sales - Lakeflow SDP Pipeline

SUBMISSION TEMPLATE

Difficulty	Marks	Tasks	Working budget
Medium	20	5	30 minutes (Medium)

Objective

Implement a file-backed Bronze/Silver/Gold Lakeflow Spark Declarative Pipeline and confirm clean category revenue.

Evidence rules - read before starting

There is NO one-task/one-screenshot mandate and NO maximum screenshot count. Provide as many clear taskwise images as the rubric requires. If both implementation and result are assessed, include BOTH. A long SQL/PySpark/pipeline definition may be split into Code_A, Code_B and Code_C screenshots, plus separate Result_A / Result_B screenshots. Images must be legible, not tiny stitched collages. A UI/configuration screenshot is acceptable when the task is menu-based. Never rerun a mutation only to capture evidence; use the original executed code and saved result. The evaluator awards marks only from the submitted visible evidence; merely supplying a workspace link is not a substitute.

Environment and required names

Use your existing assigned Unity Catalog. Reuse schema mock1 for all four questions. Create the schema if missing; do not create a catalog, storage credential, external location, or an unrelated warehouse. Each question creates ONLY its own EXTERNAL volume using the instructor-supplied complete ADLS URI beneath the approved external location. Upload source CSVs inside input/. Managed tables and SDP output must be in schema-managed Delta storage, not inside volume files.

Item	Required value
Catalog	Existing assigned trainee catalog
Schema	mock1
Question external volume	m1_m01_files
Assigned relative ADLS path	mock1/m01/files/ under trainee root
Input directory	/Volumes/<trainee_catalog>/mock1/m1_m01_files/input/
Required saved object(s)	m1_m01_retail_pipeline, m1_m01_bronze_sales, m1_m01_silver_sales, m1_m01_gold_category_sales

Key terms and formulas

- Bronze preserves the complete raw file.
- Silver retains positive quantity and positive price and adds sales_amount.
- Gold aggregates SUM(sales_amount) per category.

Dataset description

File(s): retail_sales.csv. Header present; use the supplied data types. Blank values, if any, represent missing values. Do not rename files.

Column	Type	Meaning
sale_id	STRING	Sale identifier
category	STRING	Retail category
quantity	INTEGER	Units
unit_price	DECIMAL(12,2)	Price per unit

Trainee: _____Soumiya Ayyappan_____ Candidate ID: _____ Catalog: _____

Date: _____ Notebook / SQL editor / pipeline / dashboard link: _____

Task requirements and screenshot placeholders

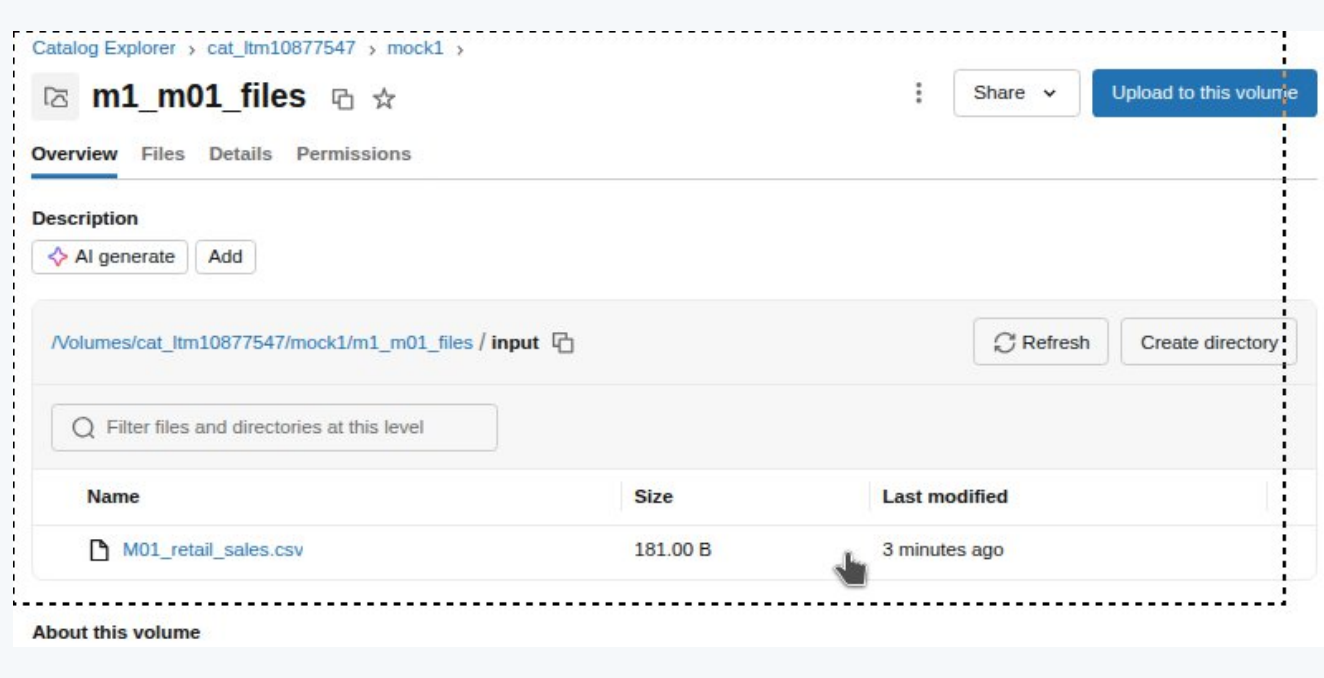
Task 1. Prepare source files and configure pipeline | 4 marks

A. Setup UI: assigned EXTERNAL volume location and file.

B. Pipeline UI: destination catalog/schema, triggered mode and attached source file.

Use descriptive filenames such as M1_M01_T01_Code_A.png, M1_M01_T01_Result_A.png, or M1_M01_T01_UI_A.png. Add _B or _C if code/output does not fit. These are examples, not a maximum number.

Paste screenshot(s) - Setup UI: _____

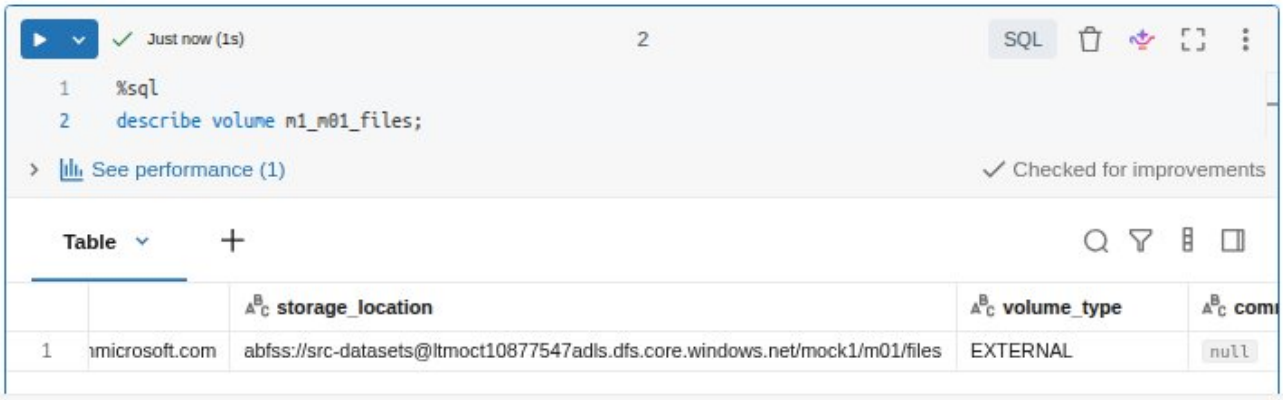


	current_catalog()	current_schema()
1	cat_ltm10877547	mock1

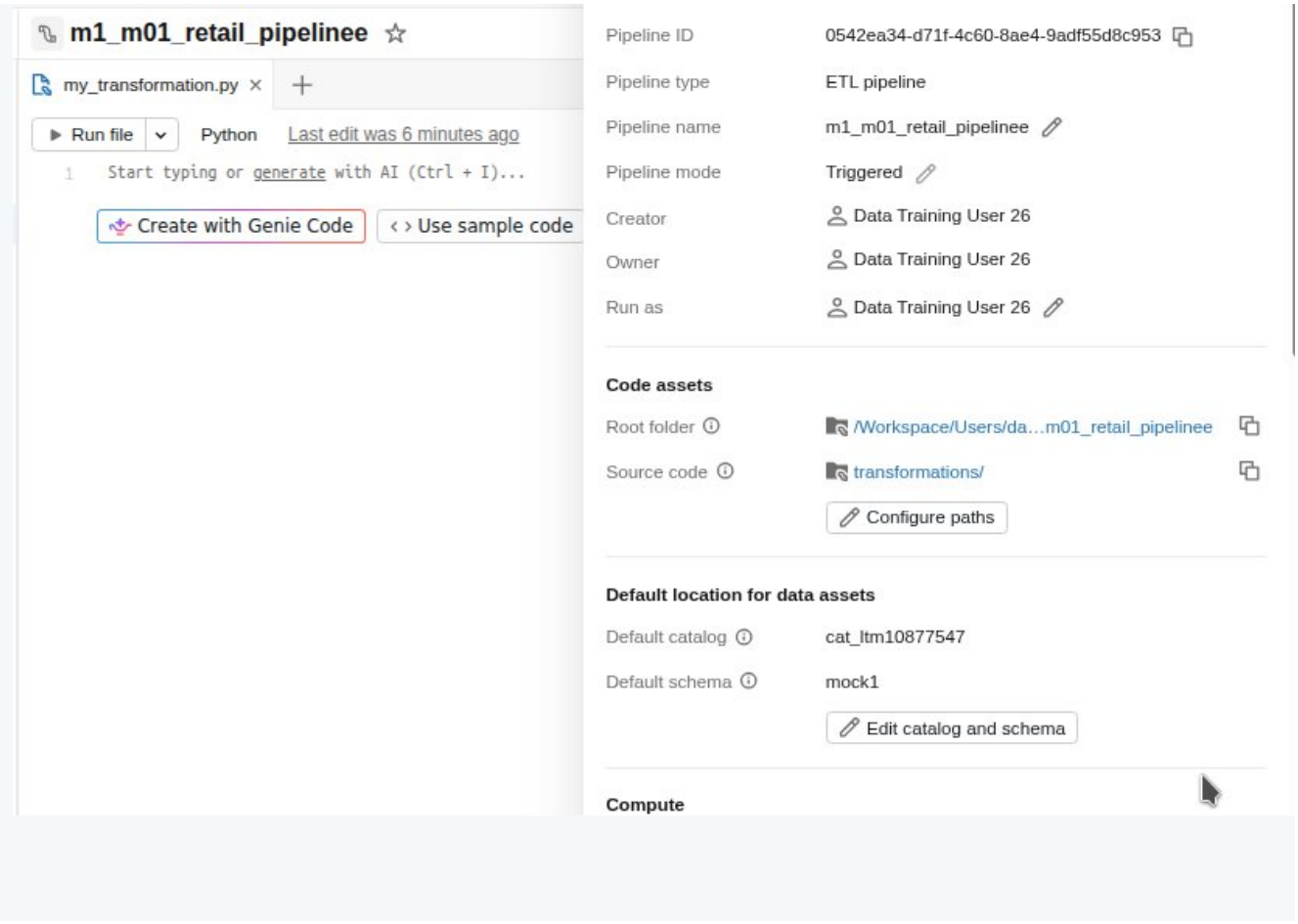
1 row | 6.672s runtime

Refreshed at 10/28/26, 5:34 AM (Now)

This result is stored as `_sqldf` and can be used in other Python and SQL cells.



Paste screenshot(s) - Pipeline UI: _____



Additional screenshot(s) for this task (if required): _____

Evidence reference / saved object: _____

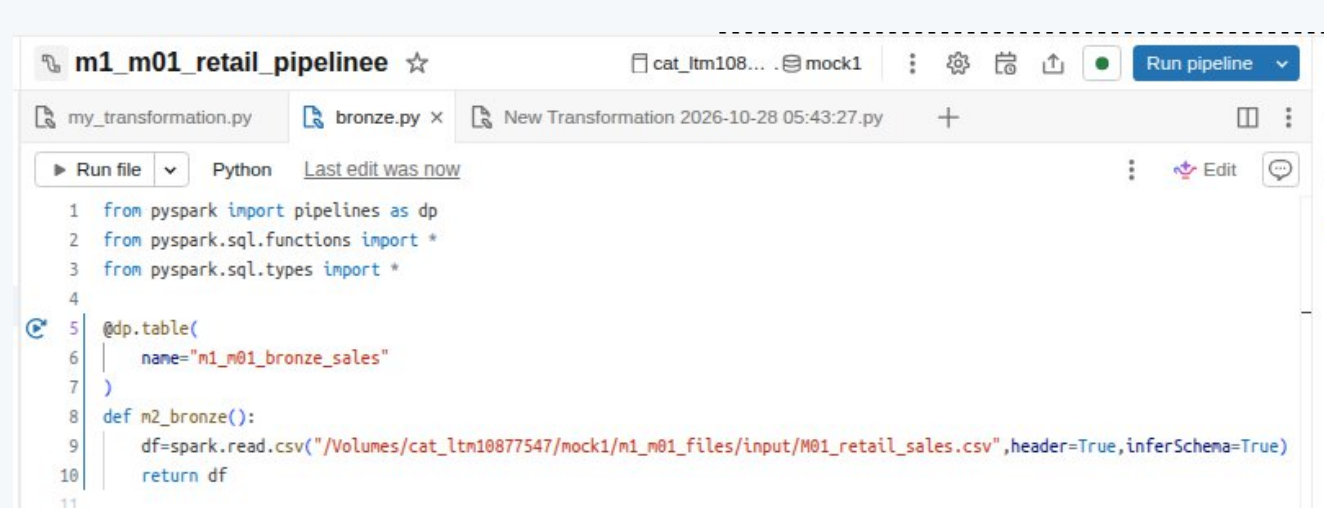
Task 2. Define Bronze ingestion | 4 marks

A. Code: complete Bronze table definition and volume source path; split long code across images.

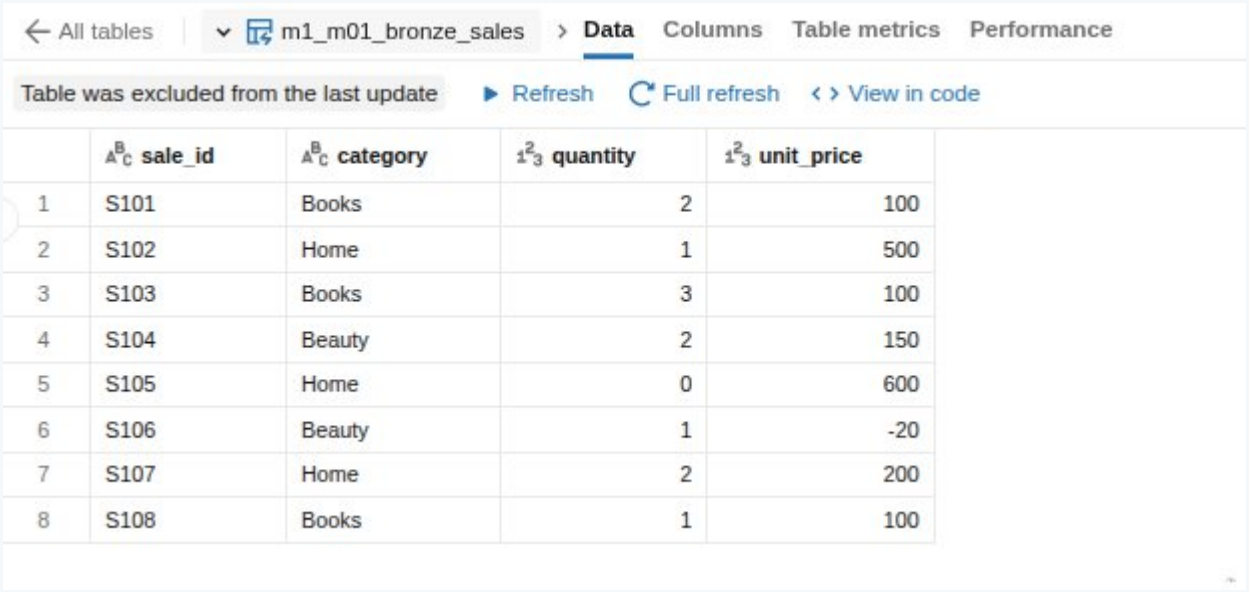
B. After successful run, result: Bronze row count eight and original four fields.

Use descriptive filenames such as M1_M01_T02_Code_A.png, M1_M01_T02_Result_A.png, or M1_M01_T02_UI_A.png. Add _B or _C if code/output does not fit. These are examples, not a maximum number.

Paste screenshot(s) - Code: _____



Paste screenshot(s) - After successful run, result: _____



	sale_id	category	quantity	unit_price
1	S101	Books	2	100
2	S102	Home	1	500
3	S103	Books	3	100
4	S104	Beauty	2	150
5	S105	Home	0	600
6	S106	Beauty	1	-20
7	S107	Home	2	200
8	S108	Books	1	100

Additional screenshot(s) for this task (if required): _____

Evidence reference / saved object: _____

Task 3. Define valid Silver sales | 4 marks

A. Code: Bronze dependency, both validity checks, calculation and Silver target.

B. Result: six records and correct derived sales_amount column.

Use descriptive filenames such as M1_M01_T03_Code_A.png, M1_M01_T03_Result_A.png, or M1_M01_T03_UI_A.png. Add _B or _C if code/output does not fit. These are examples, not a maximum number.

Paste screenshot(s) - Code: _____

my_transformation.py | bronze.py | silver.py ×

Run file | Python

```python
from pyspark import pipelines as dp
from pyspark.sql.functions import *
from pyspark.sql.types import *
@dp.materialized_view(name="m1_m01_silver_sales")
def m2_silver():
    df=spark.read.table("m1_m01_bronze_sales")
    clean_df=df.filter((col("quantity") > 0) & (col("unit_price") > 0))
    clean_df=clean_df.withColumn("sales_amount", col("quantity") * col("unit_price"))
    return clean_df
```

Paste screenshot(s) - Result: _____

All tables | m1_m01_silver_sales | Data | Columns | Table metrics | Performance

Last update: Oct 28, 2026, 5:58:21 AM | Refresh | Full refresh | View in code

	sale_id	category	quantity	unit_price	sales_amount
1	S101	Books	2	100	200
2	S102	Home	1	500	500
3	S103	Books	3	100	300
4	S104	Beauty	2	150	300
5	S107	Home	2	200	400
6	S108	Books	1	100	100

Additional screenshot(s) for this task (if required): _____

Evidence reference / saved object: _____

Task 4. Define category-level Gold metrics | 4 marks

A. Code: complete Gold aggregation definition and Silver dependency.

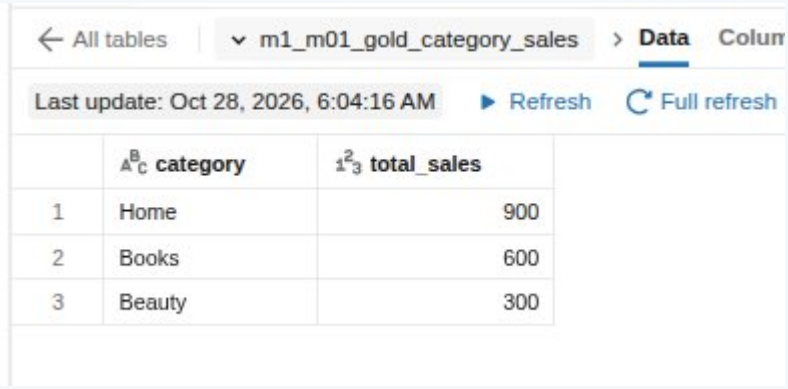
B. Result: three grouped totals and overall reconciliation.

Use descriptive filenames such as M1_M01_T04_Code_A.png, M1_M01_T04_Result_A.png, or M1_M01_T04_UI_A.png. Add _B or _C if code/output does not fit. These are examples, not a maximum number.

Paste screenshot(s) - Code: _____

```sql
CREATE OR REFRESH MATERIALIZED VIEW m1_m01_gold_category_sales
COMMENT 'Total sales amount per product category'
AS SELECT
  category,
  SUM(sales_amount) AS total_sales
FROM m1_m01_silver_sales
GROUP BY ALL;
```

Paste screenshot(s) - Result: _____



	category	total_sales
1	Home	900
2	Books	600
3	Beauty	300

Additional screenshot(s) for this task (if required): _____

Evidence reference / saved object: _____

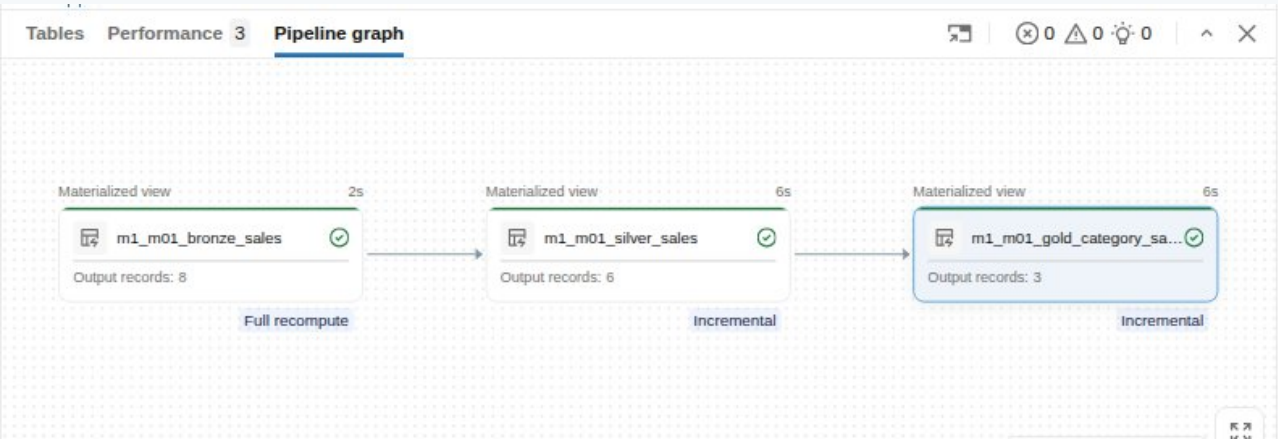
Task 5. Run and verify the pipeline | 4 marks

A. Pipeline UI: status SUCCESS/Completed, three graph nodes and destination.

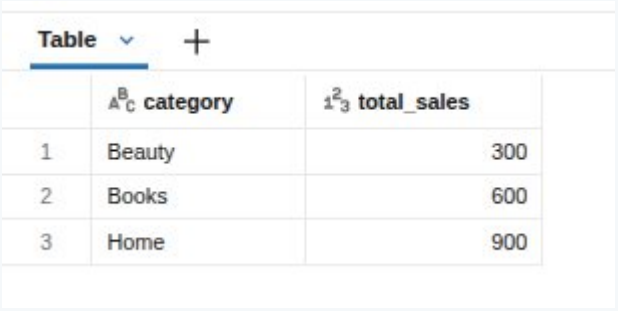
B. Results: saved Bronze/Silver row counts and Gold three-row output.

Use descriptive filenames such as M1_M01_T05_Code_A.png, M1_M01_T05_Result_A.png, or M1_M01_T05_UI_A.png. Add _B or _C if code/output does not fit. These are examples, not a maximum number.

Paste screenshot(s) - Pipeline UI: _____



Paste screenshot(s) - Results: _____



Table

	category	total_sales
1	Beauty	300
2	Books	600
3	Home	900

Additional screenshot(s) for this task (if required): _____

Evidence reference / saved object:
_____/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/m1_m01_retail_pipelinee_____
_____

Evaluation rubric and evidence map

Task	Criterion	Marks	Evidence
1	Correct governed source volume and file	2	Code / UI / Result as specified in Task 1
1	Correct triggered SDP configuration and destination	2	Code / UI / Result as specified in Task 1
2	Valid streaming Bronze definition and typed source	2	Code / UI / Result as specified in Task 2
2	All eight raw rows retained	2	Code / UI / Result as specified in Task 2
3	Correct Silver materialized view and dependency	2	Code / UI / Result as specified in Task 3
3	Both filters and sales calculation implemented	2	Code / UI / Result as specified in Task 3
4	Gold correctly depends on Silver and groups category	2	Code / UI / Result as specified in Task 4
4	Correct transaction and sales metrics	2	Code / UI / Result as specified in Task 4
5	Successful run with correct graph and destination	2	Code / UI / Result as specified in Task 5
5	Correct Bronze, Silver and Gold persisted counts	2	Code / UI / Result as specified in Task 5

Total marks: 20. Task count: 5. Screenshot count: variable; provide enough legible evidence to cover every rubric criterion.

Submission requirements

Complete the matching submission template with task-specific screenshot(s) showing every assessed code/configuration and output item. Multiple screenshots per task are allowed and sometimes necessary. Code-only evidence does not prove output; result-only evidence does not prove implementation when the rubric assesses both. Do not submit a generic solution

script in place of evidence. Keep your original notebook, Delta objects, pipeline, dashboard and version history accessible. No extra hidden tests or additional reports are required.